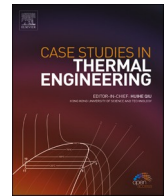




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A full-range analytical solution of the critical velocity for smoke control in tunnel fires

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ABSTRACT

The critical velocity plays a significant role in the design of the ventilation system in tunnels, from which the ventilation velocity of fresh air for preventing the smoke back-layering in tunnel fires can be determined. In this work, a full-range analytical solution to the equation for the critical velocity was derived. Specifically, we first proved that the unconstrained equation for the critical velocity has and only has one positive real solution, and then we proved that the positive real solution has a unified expression for different cases of tunnel fires. Finally, the analytical solution that satisfies the constraints was obtained, which was also compared with the iterative solution. The result shows that the analytical solution can avoid the negative and complex solutions automatically, which is more stable than the iterative solution. Furthermore, the analytical solution was compared with the simulation and experimental results, which were found to be in excellent agreements with each other. For tilted tunnels, the influence of the Froude number on the analytical solution of the critical velocity was also investigated by comparing the critical velocities obtained based on two different correlations of the Froude number.

1. Introduction

Nowadays, tunnels play more and more essential roles in public transportation and mining [1,2]. However, as a significant infrastructure hazard, tunnel fires have caused disastrous losses [3–6]. Due to the incomplete combustion of the fuel, tunnel fires generate a large amount of smoke when the fires occur. The toxic smoke would extremely threaten human lives in the tunnels, and the high-temperature smoke would highly destroy the facilities and properties in the tunnels. In the early days, longitudinal ventilation systems are usually installed in the tunnels to circulate the air in them [7]. Nowadays, ventilation systems are also employed to control the smoke movement for the protection of process safety of tunnel fires [8–10]. Commonly, the security of upstream is protected with priority. The minimum ventilation velocity for preventing the smoke back-layering is the so-called critical velocity (see Fig. 1 (a)). The ventilation under the critical velocity can not only ensure the security of the upstream, but also does not decrease many chances for the stratification of the downstream smoke. The ventilation systems in modern tunnels are usually designed according to the critical velocity.

The critical velocity was widely studied due to its importance [11–13]. Many empirical correlations were proposed to predict the critical velocity. By means of the dimensional analysis, Oka and Atkinson [14], Wu and Bakar [15], and Li et al. [11] obtained classic empirical correlations for predicting the critical velocity, respectively. These correlations are essentially piecewise functions fitted

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Nomenclature

A	Cross area of the tunnel
c_p	Specific heat capacity
Fr_c	Froude number
$f(V_c)$	The simplified function related to the critical velocity
g	Gravitational acceleration
$g(\theta)$	The simplified function related to the temperature difference
H	Tunnel height
k	The modulus of $-\frac{q}{2} + \sqrt{-\Delta} \bullet i$
k_l	The modification coefficient in the Froude number
p	Denotation of $-\dot{Q}_c^2 Fr_c / c_p^2 \rho_\infty^2 A^2 gH$
$\dot{Q}_c$	Convective heat release rate of the fire
$\dot{Q}$	Total heat release rate of the fire
$\dot{Q}_c^*$	Dimensionless convective HRR
$\dot{Q}^*$	Dimensionless total HRR
q	Denotation of $-\dot{Q}_c^2 Fr_c T_\infty / c_p^2 \rho_\infty^2 A^2 gH$
T_∞	Ambient temperature
T_f	Temperature of the fire smoke flow in the downstream
V_c	Critical ventilation velocity
$V_c^{*,i}$	The i-th root of the derivative of $f(V_c)$
V_c^s	The final solution of critical velocity of equation (4)

Greek and other symbol

γ	The tunnel slope
Δ	The discriminant
ΔV_c	Difference between critical velocities obtained with different correlations
$\Delta\rho$	Density difference between the fresh air and the smoke flow
η	Correction factor of the Froude number
θ	Temperature difference between the fresh air and the smoke flow
θ_i	The i-th root if equation (7)
$\theta^{*,i}$	The i-th root of the derivative of $g(\theta)$
ρ_∞	Ambient density of fresh air
ρ_f	Density of the fire smoke flow in the downstream
φ	The argument of $-\frac{q}{2} + \sqrt{-\Delta} \bullet i$

from experimental data. The theoretical analysis can also provide good means for obtaining the critical velocity, Thomas [16] first theoretically gave a correlation for predicting the critical velocity based on the understanding of physical phenomenon. He used a modified Froude number to build a model related to the critical velocity, then the critical velocity can be obtained by solving iteratively a set of non-linear equations (a brief review can be seen in section 2). However, Thomas did not consider the effect of the tunnel slope and did not test his results under high heat release rates (HRR). In addition, he considered only a constant Froude number which is equal to 1 in his work. Subsequently, Danziger and Kennedy [17,18] modified Thomas' model, in which the Froude number was related to the tunnel slope. In 2017, a further modification to Danziger and Kennedy's model was provided in the NFPA 502 Standard, in which the Froude number is also related to the HRR. Recently, Li, et al. [19] conducted a set of experiments and simulations, and they modified part of the Froude number model that is related to the tunnel slope. These equations show good performance in predicting the critical velocity. However, since the equations are non-linear, they were usually solved iteratively, from which no explicit mathematical expression of the critical velocity can be obtained [20]. On the other hand, obtaining an analytical solution to the critical

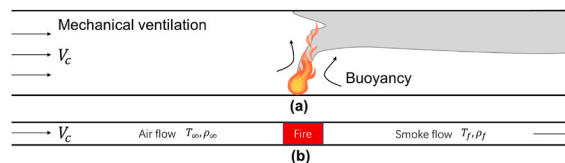


Fig. 1. (a) Fire scenario and (b) the network model.

velocity is very useful not only for the purposes of data verification and validation, but also for guiding the development of mathematical and computer models.

Tarada and Crawley [21] firstly tried to give an analytical solution to Thomas' equations. He simplified the equations into a cubic equation. However, only the simplest situation was considered in his work: the cubic equation has only one real root and the other two are complex roots (i.e., the discriminant is larger than zero). Yousefi et al. [22] applied Tarada's solution to predict the critical velocity. However, Both Tarada and Yousefi did not prove the unique real root of the cubic equation is positive, so the unique real root may be a negative number which contradicts the assumptions used in deriving the equations. In addition, the given expression is not simplified, which has too many intermediate parameters. Moreover, their solution is only applicable for the case where the discriminant Δ is larger than zero, which makes its application limited. Therefore, the motivation of the paper is to overcome these flaws and to give a unified expression that applies in full range.

In this study, we derived an analytical solution that applies to all cases (any HRR and tunnel parameters). Firstly, we proved the cubic equation has only one positive root for any parameters. Next, we proved the analytical solution has a unified expression under all of the three situations (the discriminant $\Delta <, =, \text{ and } > 0$). Furthermore, the final analytical expression of the critical velocity was obtained. The analytical solution was also compared with the iterative, the simulation and the experimental results. Excellent agreement was shown. The influence of the Froude number in different models on the critical velocity was discussed as well.

It is stressed that this work is interested in the analytical solution based on the correlation proposed by Kennedy [17] and the comparison with other correlations is considered beyond the scope of this article. Nevertheless, this study will illustrate the potential of the proposed analytical solution to the equations governing the critical velocity that have been solved numerically for decades.

2. Review of the critical velocity equations

Fig. 1 (a) shows the tunnel fire scenario, the smoke stops at the fire source under the action of ventilation (with critical velocity) and buoyancy. In 1968, by neglecting the molecular diffusion and viscosity and, in the ideal case, heat loss and friction at the walls, Thomas modelled the fire scenario that the smoke movement is controlled by the ventilation velocity and the buoyancy induced by the high temperature, based on the network model [16], as shown in Fig. 1 (b). The velocity head of the fresh airflow can be written as $\rho_\infty V_c^2$, and the buoyancy head of the smoke flow can be written as $\Delta\rho gH$. Then the critical state that the smoke just stops at the fire source is determined by the Froude number, which is defined as the buoyancy head divided by the velocity head [23]:

$$\text{Fr}_c = \frac{\Delta\rho gH}{\rho_\infty V_c^2}, \quad (1)$$

where Fr_c is the critical Froude number, $\Delta\rho$ is the density difference between the fresh air and the smoke flow, g is the gravitational acceleration, H is the tunnel height, ρ_∞ is the density of ambient fresh air, and V_c is the critical ventilation velocity. The Froude number defined in Eq. (1) was first modified by Thomas [16]. By assuming the air and the smoke flow are ideal gases, the following expression can be obtained:

$$\Delta\rho = \rho_\infty - \rho_f = \rho_\infty \left(1 - \frac{T_\infty}{T_f}\right) = \rho_\infty \left(\frac{\theta}{T_f}\right), \quad (2)$$

where ρ_f is the density of the fire smoke flow in the downstream, T_∞ is the ambient temperature, T_f is the temperature of the fire smoke flow in the downstream, and $\theta = T_f - T_\infty$. From the conservation of energy, we have:

$$\dot{Q}_c = c_p \rho_\infty A V_c \theta, \quad (3)$$

where $\dot{Q}_c$ is the convective heat release rate of the fire, c_p is the specific heat of air, and A is the cross-sectional area of the tunnel.

Finally, the equations related with the critical velocity can be obtained by simplifying Eqs. (1)–(3) [17]:

$$\left\{ \begin{array}{l} V_c = \left(\frac{\dot{Q}_c g H}{c_p \rho_\infty A (\theta + T_\infty) \text{Fr}_c} \right)^{\frac{1}{3}} \\ \theta = \frac{\dot{Q}_c}{c_p \rho_\infty A V_c} \\ \text{Constraints : } \begin{cases} V_c > 0 \\ \theta > 0 \end{cases} \end{array} \right. \quad (4)$$

It is noted that Eq. (4) has default constraints in the derivation process, that is $V_c > 0$ and $\theta > 0$. So far, Eq. (4) can be solved by iterative calculation in the full range of parameters. The computer software called the Subway Environmental Simulation (SES) was

developed to iteratively calculate the critical velocity [18]. However, the significance of the iterative solution is far inferior to the analytical solution given that it can be obtained.

3. Derivation and proof of the analytical solution

If the constraints are neglected, Eq. (4) can be reduced into a single equation that is only related to the critical velocity V_c by eliminating θ :

$$(\text{Fr}_c c_p \rho_\infty A T_\infty) \bullet V_c^3 + (\text{Fr}_c \dot{Q}_c) \bullet V_c^2 - g H \dot{Q}_c = 0. \quad (5)$$

Eq. (5) is a unary cubic equation which means it has three roots. However, in the previous studies [17,18], Eq. (4) was considered to have only one solution by default: the other two solutions were neglected without any discussion (i.e., more real roots may exist). Yousefi et al. [22] considered a special situation of Eq. (5): it has a unique real root and two complex roots by setting the discriminant of Eq. (5) larger than zero. Even so, they did not prove the real root is always positive so that the constraints in Eq. (4) are satisfied. In this section, we will prove that Eq. (4) has only one real root and it is positive always. Based on this, we will give an exact analytical solution to Eq. (4) that applies to the full range and prove it.

3.1. Proof of Eq. (4) having only one positive real root

In order to prove Eq. (5) has a unique real solution, we first prove Eq. (5) has and only has one positive root. Since Eq. (5) does not have constraints, if the above proposition is proved, then the only positive root of Eq. (5) can be considered as the final solution of Eq. (4) (with constraints).

Proof:

Let

$$f(V_c) = (\text{Fr}_c c_p \rho_\infty A T_\infty) \bullet V_c^3 + (\text{Fr}_c \dot{Q}_c) \bullet V_c^2 - g H \dot{Q}_c.$$

We have: $f(0) = -g H \dot{Q}_c < 0$; and $\lim_{V_c \rightarrow +\infty} f(V_c) = +\infty > 0$. As $f(V_c)$ is a continuous function, it is known that there exists at least one positive real V_c that satisfies $f(V_c) = 0$, according to the intermediate value theorem [24].

Next, we prove that for all values of V_c that satisfies $f(V_c) = 0$, there is only one positive real value of V_c . Let

$$\frac{df(V_c)}{dV_c} = (3\text{Fr}_c c_p \rho_\infty A T_\infty) \bullet V_c^2 + (2\text{Fr}_c \dot{Q}_c) \bullet V_c \equiv 0, \quad (6)$$

then we have its roots $V_c^{*,1} = 0$, and $V_c^{*,2} = \frac{-2\dot{Q}_c}{3c_p \rho_\infty A T_\infty} < 0$.

Since Eq. (6) is a quadratic function and $3\text{Fr}_c c_p \rho_\infty A T_\infty > 0$, it is implied that $f(V_c)$ monotonically increases from $-\infty$ to $V_c^{*,2}$, then monotonically decreases from $V_c^{*,2}$ to $V_c^{*,1}$ (equal to zero), and finally monotonically increases from $V_c^{*,1}$ to $+\infty$. Hence, $f(V_c)$ has and only has one positive root between 0 to $+\infty$. ■

The possible shapes of $f(V_c)$ is shown schematically in Fig. 2. The value of $f(V_c^{*,2})$ determines the characteristics of another two roots of Eq. (5):

- a) Two negative real roots and one positive real root, if $f(V_c^{*,2}) > 0$;
- b) Two negative real roots (multiple roots) and one positive real root, if $f(V_c^{*,2}) = 0$;
- c) Two complex roots and one positive real root, if $f(V_c^{*,2}) < 0$.

3.2. Determination of the analytical solution of Eq. (4)

As Eq. (5) is a cubic equation, it has three theoretical solutions. However, the three roots are very complicated, at least, by observing the expression. It is not easy to determine whether they are positive or negative, and real or complex numbers [25]. Nevertheless, the three complicated roots still need to be calculated first, then the unique positive real root of Eq. (4) can be obtained after discarding the negative real or the complex roots of Eq. (5).

In section 3.1, we know the roots of Eq. (5) may be divided into three cases a), b), and c), so the positive real root may need to be obtained by comparing the expressions of the roots in each case. Fortunately, it is found that this positive real root has a unique expression for the three cases a), b), and c). In this section, the unique expression will be proved and derived. In the following, we deal with equations related to θ instead of V_c , because the roots of the equations related to θ are more compact and therefore easier to deal with.

Proof:

The following equation that is only related to θ can be derived by eliminating V_c in Eq. (4):

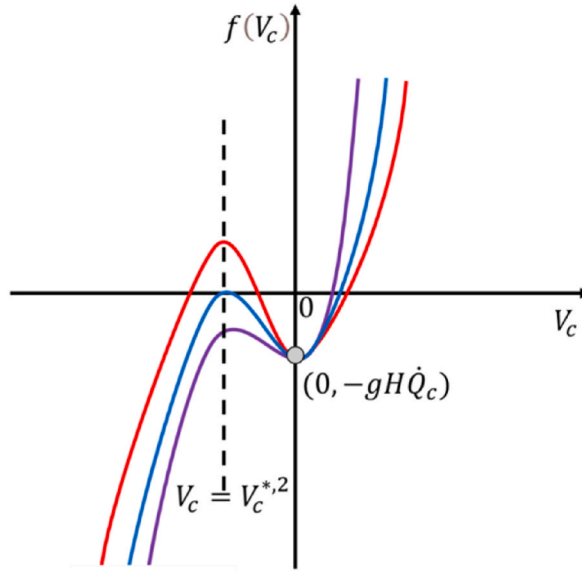


Fig. 2. The possible shapes of $f(V_c)$. The red, blue, and purple curves represent the situations of a), b) and c), respectively.

$$\theta^3 - \frac{\dot{Q}_c^2 \text{Fr}_c}{c_p^2 \rho_\infty^2 A^2 gH} \theta - \frac{\dot{Q}_c^2 \text{Fr}_c T_\infty}{c_p^2 \rho_\infty^2 A^2 gH} = 0 \quad (7)$$

Denoting $p = -\frac{\dot{Q}_c^2 \text{Fr}_c}{c_p^2 \rho_\infty^2 A^2 gH}$, $q = -\frac{\dot{Q}_c^2 \text{Fr}_c T_\infty}{c_p^2 \rho_\infty^2 A^2 gH}$, then it is obvious that $p < 0$ and $q = pT_\infty < 0$. Eq. (7) is rewritten as

$$\theta^3 + p\theta + q = 0 \quad (8)$$

Unlike Eq. (5), Eq. (8) does not have the quadratic term. This leads an advantage: the expressions of the roots of Eq. (8) are quite concise [25], which are of the form:

$$\begin{cases} \theta_1 = w_1 \cdot \sqrt[3]{\frac{-q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + w_1 \cdot \sqrt[3]{\frac{-q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} \\ \theta_2 = w_2 \cdot \sqrt[3]{\frac{-q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + w_3 \cdot \sqrt[3]{\frac{-q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} \\ \theta_3 = w_3 \cdot \sqrt[3]{\frac{-q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + w_2 \cdot \sqrt[3]{\frac{-q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} \end{cases} \quad (9)$$

where w_1, w_2, w_3 are respectively equal to 1, $e^{i\frac{2\pi}{3}}$, $e^{-i\frac{2\pi}{3}}$ (three primitive cube roots of unity), respectively.

Back to Eq. (4), it is known that $\theta > 0$ when $V_c > 0$ and vice versa. It is also known that θ is a complex number when V_c is a complex number, whereas θ is a real number when V_c is a real number. This means the unique positive real root of Eq. (5) is found when the positive real root of Eq. (8) is found. This leads us to find the positive real root of Eq. (8). Actually, it can be proved that Eq. (8) has a unique positive real root as well, following the same proof process as in section 3.1. Let

$$g(\theta) = \theta^3 + p\theta + q.$$

It is noted that (1) $g(0) = q < 0$; and (2) $\theta^{*,1} = \sqrt[3]{\frac{-p}{3}}$ and $\theta^{*,2} = -\sqrt[3]{\frac{-p}{3}}$ are the roots of equation $\frac{dg(\theta)}{d\theta} = 3\theta^2 + p = 0$. So the possible shapes of $g(\theta)$ is schematically shown in Fig. 3.

Denoting $\Delta = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3$, we can know $\Delta \geq 0$ are equivalent to $g(\theta^{*,2}) \leq 0$, and vice versa. The following three cases are considered:

Case A. $\Delta > 0$ ($g(\theta^{*,2}) < 0$)

As $q < 0$, then $\frac{-q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3} = \frac{-q}{2} + \sqrt{\Delta} > 0$. As $p < 0$ and $q < 0$, then $\frac{-q}{2} = \sqrt{\left(\frac{q}{2}\right)^2} > \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}$, i.e., $\frac{-q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3} > 0$. As a result, $\theta_1 > 0$ is a positive real number. In addition, $p \neq 0$, $\frac{-q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3} \neq \frac{-q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}$, so both θ_2 and θ_3 are complex

numbers. This is because the combination of two unequal real numbers rotated by $\frac{2}{3}\pi$ and $-\frac{2}{3}\pi$ in the complex plane must be a complex number, as shown in Fig. 4.

Case B. $\Delta = 0$ ($g(\theta^{*,2}) = 0$)

As $\Delta = 0$, it is easy to know $\theta_1 = 2\sqrt[3]{\frac{-q}{2}} > 0$. Following similar analysis to Case A, then it is obtained $\theta_2 = \theta_3 = -2\sqrt[3]{\frac{-3q}{2}} < 0$.

Case C. $\Delta < 0$ ($g(\theta^{*,2}) > 0$)

As $\Delta < 0$, thus $\sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3} = \sqrt{-\Delta} \bullet i$. Next, $\theta_1 = \sqrt[3]{\frac{-q}{2} + \sqrt{-\Delta} \bullet i} + \sqrt[3]{\frac{-q}{2} - \sqrt{-\Delta} \bullet i}$. Since $\frac{-q}{2} > 0$, we can denote $\frac{-q}{2} + \sqrt{-\Delta} \bullet i$ as $k \bullet e^{i\varphi}$, where $k = \sqrt{\frac{q^2}{4} - \Delta}$ and $\varphi = \arctan\left(-\frac{2\sqrt{-\Delta}}{q}\right)$. Here, because $\frac{-q}{2} > 0$ and $\sqrt{-\Delta} > 0$, it is known $k > 0$ and $0 < \varphi < \frac{\pi}{2}$. In addition, $\frac{-q}{2} - \sqrt{-\Delta} \bullet i = k \bullet e^{-i\varphi}$ is the complex conjugate of $\frac{-q}{2} + \sqrt{-\Delta} \bullet i$. As a result, $\theta_1 = \sqrt[3]{k \bullet e^{i\frac{\varphi}{3}}} + \sqrt[3]{k \bullet e^{-i\frac{\varphi}{3}}} = 2\sqrt[3]{k} \cos\left(\frac{\varphi}{3}\right) > 0$, where $0 < \frac{\varphi}{3} < \frac{\pi}{6}$. Similarly, $\theta_2 = \sqrt[3]{k \bullet e^{i(\frac{\varphi}{3} + \frac{2\pi}{3})}} + \sqrt[3]{k \bullet e^{-i(\frac{\varphi}{3} + \frac{2\pi}{3})}} = 2\sqrt[3]{k} \cos\left(\frac{\varphi}{3} + \frac{2\pi}{3}\right)$, and $\theta_3 = \sqrt[3]{k \bullet e^{i(\frac{\varphi}{3} - \frac{2\pi}{3})}} + \sqrt[3]{k \bullet e^{-i(\frac{\varphi}{3} - \frac{2\pi}{3})}} = 2\sqrt[3]{k} \cos\left(\frac{\varphi}{3} - \frac{2\pi}{3}\right)$. Due to the fact that $\frac{2\pi}{3} < \frac{\varphi}{3} + \frac{2\pi}{3} < \frac{5\pi}{6}$ and $-\frac{2\pi}{3} < \frac{\varphi}{3} - \frac{2\pi}{3} < -\frac{\pi}{2}$, it is known $\theta_2 < 0$ and $\theta_3 < 0$ (Fig. 5).

In summary, $\theta_1 > 0$ holds for all cases, the solution of θ in Eq. (4) thus has a unified expression as:

$$\theta^s = \sqrt[3]{\frac{\dot{Q}_c^2 \text{Fr}_c T_\infty}{2c_p^2 \rho_\infty^2 A^2 gH}} + \sqrt[3]{\left(\frac{\dot{Q}_c^2 \text{Fr}_c T_\infty}{2c_p^2 \rho_\infty^2 A^2 gH}\right)^2 - \left(\frac{\dot{Q}_c^2 \text{Fr}_c}{3c_p^2 \rho_\infty^2 A^2 gH}\right)^3} \\ + \sqrt[3]{\frac{\dot{Q}_c^2 \text{Fr}_c T_\infty}{c_p^2 \rho_\infty^2 A^2 gH} - \sqrt[3]{\left(\frac{\dot{Q}_c^2 \text{Fr}_c}{2c_p^2 \rho_\infty^2 A^2 gH}\right)^2 - \left(\frac{\dot{Q}_c^2 \text{Fr}_c}{3c_p^2 \rho_\infty^2 A^2 gH}\right)^3}}. \quad (10)$$

Finally, the solution of V_c in Eq. (4) has a unified expression as:

$$V_c^s = \left(\frac{\dot{Q}_c gH}{c_p \rho_\infty A \text{Fr}_c}\right)^{\frac{1}{3}} \bullet \left(\sqrt[3]{\frac{\dot{Q}_c^2 \text{Fr}_c T_\infty}{2c_p^2 \rho_\infty^2 A^2 gH}} + \sqrt[3]{\left(\frac{\dot{Q}_c^2 \text{Fr}_c T_\infty}{2c_p^2 \rho_\infty^2 A^2 gH}\right)^2 - \left(\frac{\dot{Q}_c^2 \text{Fr}_c}{3c_p^2 \rho_\infty^2 A^2 gH}\right)^3} \right. \\ \left. + \sqrt[3]{\frac{\dot{Q}_c^2 \text{Fr}_c T_\infty}{c_p^2 \rho_\infty^2 A^2 gH} - \sqrt[3]{\left(\frac{\dot{Q}_c^2 \text{Fr}_c}{2c_p^2 \rho_\infty^2 A^2 gH}\right)^2 - \left(\frac{\dot{Q}_c^2 \text{Fr}_c}{3c_p^2 \rho_\infty^2 A^2 gH}\right)^3}} + T_\infty \right)^{-\frac{1}{3}}. \quad (11)$$

(V_c^s, θ^s) is the theoretical solution set of Eq. (4).

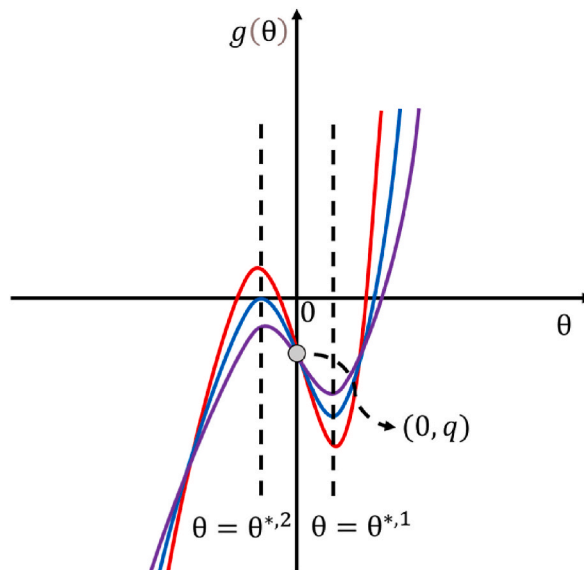
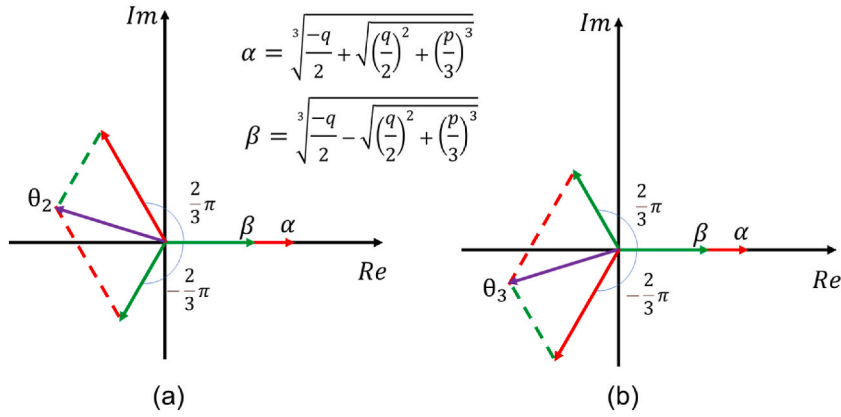
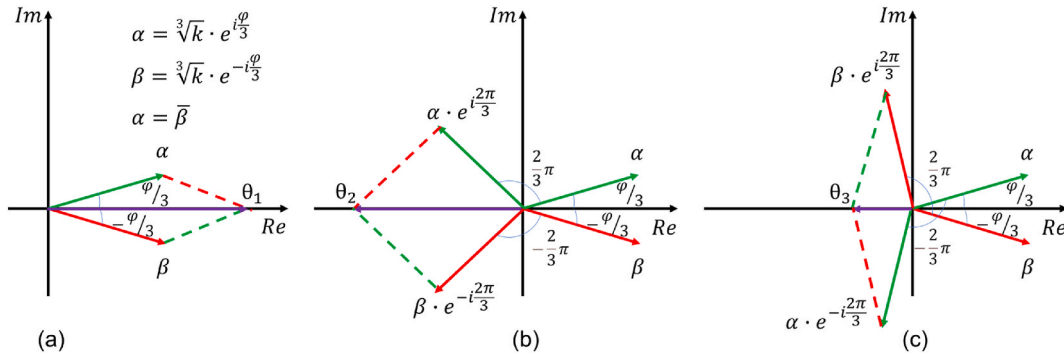


Fig. 3. The possible shapes of $g(\theta)$. The red, blue, and purple curves represent the situations of a), b) and c), respectively.

Fig. 4. Schematic of θ_2 and θ_3 for the case $\Delta > 0$.Fig. 5. Schematic of (a) θ_1 , (b) θ_2 , and (c) θ_3 for the case $\Delta < 0$.

4. Applications and case studies

4.1. Comparison with the iterative solution

On the basis of summarizing many experiment results, the NFPA 502 [26] provided a recommended Froude number for predicting the critical velocity as:

$$Fr_c = (k_l \cdot (1 + 0.0374 \cdot \gamma^{0.8}))^{-3}, \quad (12)$$

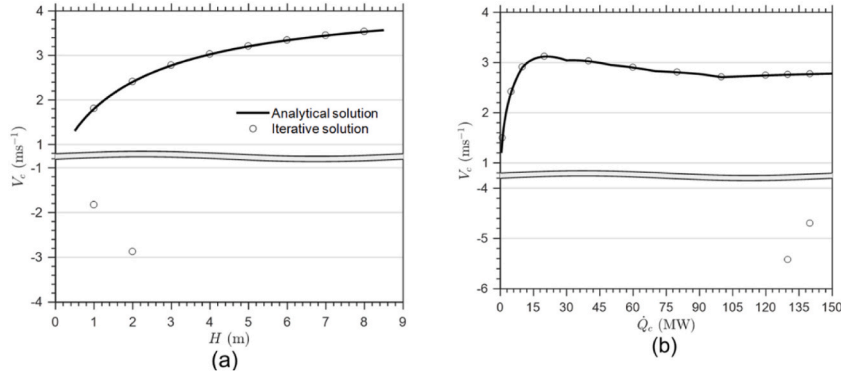
where k_l is the modification coefficient in the Froude number (a piecewise function related to $\dot{Q}_c$, as shown in Table 1) and γ is the tunnel slope. It is noted Table 1 needs to be used with linear interpolation. In this comparison, the tunnel height, the tunnel width, the ambient temperature, and the tunnel slope are set as 4 m, 5 m, 25 °C, and 0%, respectively.

Fig. 6 (a) shows the comparison between the analytical solution and the iterative solution for the various tunnel height, in which the $\dot{Q}_c$ is fixed as 40 MW. Due to the increase of the tunnel height, the buoyancy force of the smoke flow ($\Delta\rho gH$) increases as well. The Froude number is constant when the $\dot{Q}_c$ is fixed, the critical velocity thus increases with the tunnel height. It is seen that the analytical solution agrees well with the iterative solution. However, erroneous results may be obtained with the iterative solution for low-height tunnels if the initial value was not selected appropriately, as shown in the figure with the negative values of V_c . Actually, as the $\dot{Q}_c$ is fixed in this comparison, $\Delta < 0$ (case C) is equivalent to $H < \frac{4\dot{Q}_c^2 Fr_c}{27c_p^2 \rho_\infty^2 A^2 T_\infty^2 g}$. Hence, for low-height tunnels, the iterative solution may return negative values as the iterative method does not consider that V_c and θ are larger than zero by default.

Fig. 6 (b) displays the critical velocity as a function of convective HRR, obtained from both the analytical solution and iterative solution. The tunnel height is fixed as 4 m here. Apparently, the average buoyancy force of the smoke flow ($\Delta\rho gH$) increases with the convective HRR, so the inertial force ($\rho_\infty V_c^2$) should increase with the convective HRR as well. Therefore, the critical velocity increases with the convective HRR. However, as the Froude number is modified according to the k_l value listed in Table 1 (decrease with $\dot{Q}_c$), the critical velocity is no longer monotonically increasing with the convective HRR after the convective HRR is larger than 10 MW. Similar to Fig. 6 (a), the analytical solution is overall in excellent agreement with the iterative solution. But for high $\dot{Q}_c$, i.e., when $\dot{Q}_c > \frac{c_p \rho_\infty A T_\infty}{2} \sqrt{\frac{27gH}{Fr_c}}$, $\Delta < 0$ then. In this situation, as Fig. 2 shows that Eq. (5) has negative solutions. Therefore, the iterative solution may

Table 1A range of values for k_l under various HRRs – NFPA 502, 2017.

$\dot{Q}_c$ (MW)	k_l
≥ 100	0.606
90	0.62
70	0.64
50	0.68
30	0.74
≤ 10	0.87

**Fig. 6.** The comparison of the critical velocity obtained with the analytical solution and the iterative solution for (a) different slopes and different convective HRR.

return a negative value (depending on the initial value) when $\dot{Q}_c$ is large enough.

Overall, the proposed analytical solution can agree well with the positive iterative solution, which validates the correctness of the analytical solution. In addition, the analytical solution automatically keeps the positive solution, which avoids the negative solution and complex solution. As a result, the analytical solution applies to the full range of the various tunnel height, convective HRR, etc.

4.2. Comparisons with the simulation and experimental results

Li et al. [19] experimentally and numerically studied the effect of the tunnel slope on the critical velocity in tilted tunnels. In their investigation, the total HRR was used for predicting the critical velocity, i.e., the convective HRR $\dot{Q}_c$ was replaced by the total HRR $\dot{Q}$ in Eq. (4). In addition, they proposed a modified correlation of the Froude number as

$$Fr_c = (0.606 \cdot (1 + 0.065 \cdot \gamma))^{-3}. \quad (13)$$

Fig. 7 shows the comparison between the analytical solution and the simulation results, as well as the experimental results. The simulation and experiment were conducted using a tunnel with fixed length of 200 m, width of 8.5 m and height of 5 m. The ambient temperature is 25 °C. The total HRR varies from 5 MW to 18 MW, and the tunnel slope varies from 3% to 7%. The simulation results were carried out by the Fire Dynamics Simulator (FDS) version 6.0. The experiments were conducted by a 1:8 scale model, in which the HRR was calculated by fuel supply flow rate multiplied by the heat of combustion of the fuel. The thermo-anemometer was used to measure the longitudinal ventilation velocity. More detail can be found in Ref. [19].

The analytical solution overall agrees well with the simulation results and the experimental results. The agreement is better for higher tunnel slope. This is because the correlation of the Froude number fits better the experimental data for higher tunnel slope, as shown in Ref. [19]. Since the tunnel slope strengthens the buoyancy of the smoke flow, the critical velocity overall increases with the tunnel slope.

The results show that the analytical solution can work well for predicting the critical velocity, especially it can provide a complete mathematical expression rather than discrete data points obtained with the iterative solution. Nevertheless, it is seen the analytical solution is related to the Froude number, so one should carefully test the correlations of the Froude number when applying it.

4.3. The influence of the Froude number correlation on the analytical solution: considering tunnel slope and HRR

As discussed above, for tilted tunnels, the correlation of the Froude number recommended in NFPA 502 [26] is different from that recommended by Li et al. [19]. The difference between them needs to be evaluated using the analytical solution. In the comparison, the tunnel height, tunnel width, and ambient temperature are 4 m, 5 m, and 25 °C, respectively. The convective HRR varies from 0.5 MW

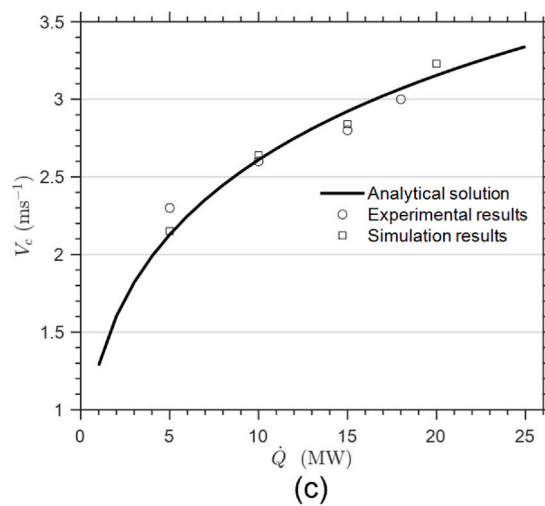
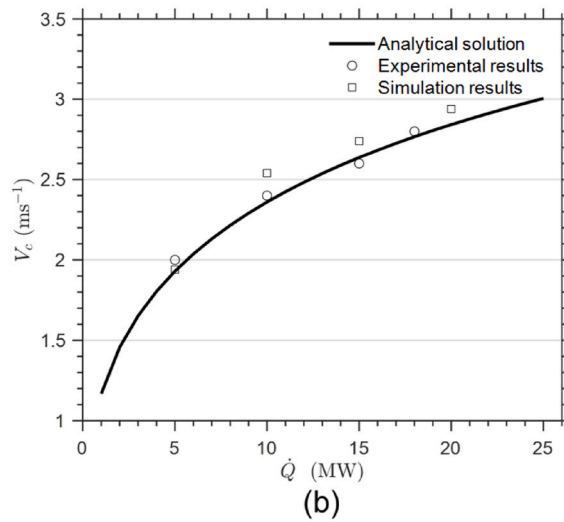
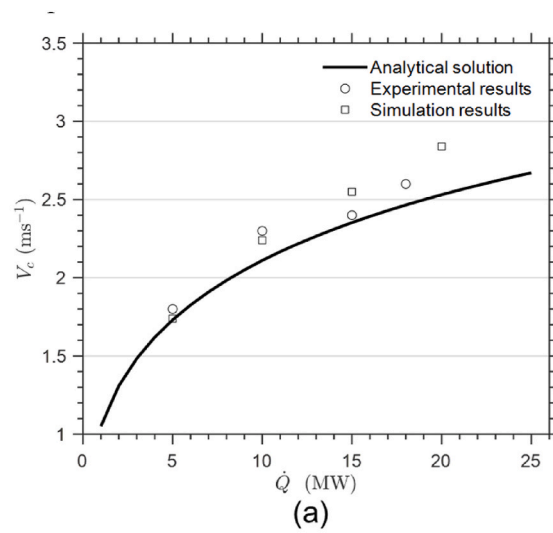


Fig. 7. Comparison of the critical velocity obtained with the analytical solution, experimental results, and simulation results for different tunnel slopes: (a) 3%, (b) 5%, (c) 7%.

to 150 MW, and the tunnel slope varies from 0% to 10%. The radiation fraction is assumed to be 30%, which means the convective HRR $\dot{Q}_c$ accounts for 70% of the total HRR $\dot{Q}$.

Due to the fact that the expression of the analytical solution is complicated, it is not easy to see directly how it changes with variables such as the tunnel slope γ and $\dot{Q}_c$. Fig. 8 displays the results that are calculated respectively based on the correlations recommended by NFPA 502 [26] and Li et al. [19]. The critical velocity obtained based on the correlation recommended by Li et al. [19] increases with both the tunnel slope and the HRR. This is also confirmed by the blue and the black curves in Fig. 8 (b). The critical velocity obtained with the correlation recommended by NFPA also increases with the tunnel slope, whereas it does not monotonically increase with the HRR, as shown in Fig. 8(a). When the slope is fixed and under the other specified parameters, the highest critical velocity is found for the HRR in the range of 20–25 MW (note: not a constant number). The correlation recommended by NFPA limits the increases of the critical velocity when the HRR is high, which is more in line with the real scene. As also mentioned by Li et al. [11], the critical velocity is almost a constant when the HRR is high enough. Overall, the critical velocity increases sharply with the HRR when the HRR is low (less than 25 MW in this study) for both correlations. However, the correlation recommended in NFPA 502 (2017) is more in line with the real scene for high HRR.

Fig. 9 shows the difference of V_c calculated based on the correlations proposed by Li et al. and by NFPA. The black curve in Fig. 9 is the contour of zero difference. It is seen that the critical velocity calculated with the correlation by NFPA is higher than that calculated with the correlation by Li et al. when HRR is less than 90 MW and when the tunnel slope is equal to zero. However, with the increase of the tunnel slope, the critical velocity calculated with the correlation by Li et al. can exceed that obtained with the correlation by NFPA at lower HRR. This means that the influence of the tunnel slope on the critical velocity is greater when using the correlation by Li et al. than that from NFPA 502. In fact, since the difference comes from applying different correlations of the Froude number in calculating the critical velocity based on Eq. (4), one only needs to consider $\frac{\partial Fr}{\partial \gamma}$ when evaluating the effect of the tunnel slope on the critical velocity, which is expressed respectively as $-0.0898k_i^{-3}(1 + 0.0374\gamma^{0.8})^{-4}\gamma^{-0.2}$ and $-0.876(1 + 0.065\gamma)^{-4}$ based on the correlations from NFPA 502 and by Li et al. It is easy to show that, in the specified range of γ , $\frac{\partial Fr}{\partial \gamma}$ is more significant if the correlation by Li et al. is considered.

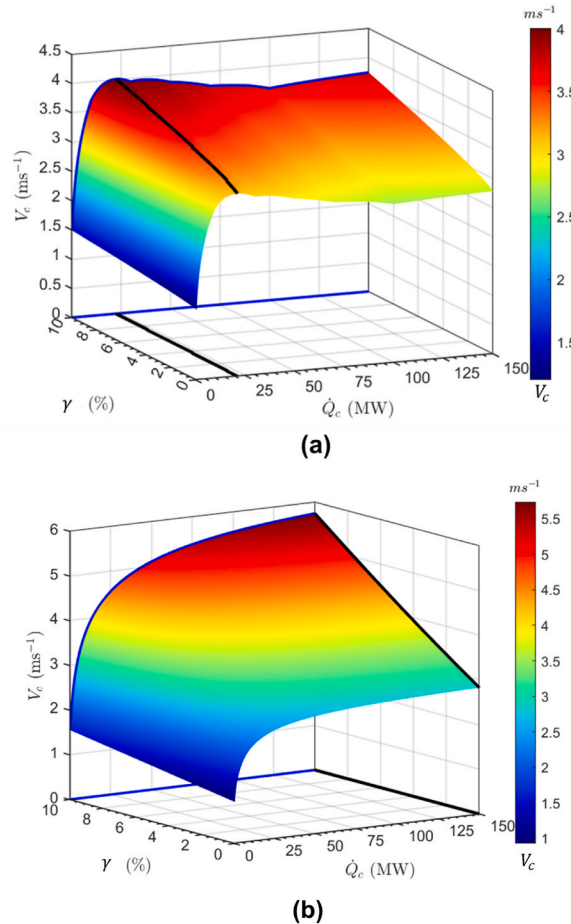


Fig. 8. The analytical solutions with the correlation of the Froude number recommended by (a) NFPA 502 [26] and (b) Li et al. [11]. The blue curves are the maximum values when the tunnel slope is fixed. The black curves are the maximum values when $\dot{Q}_c$ is fixed.

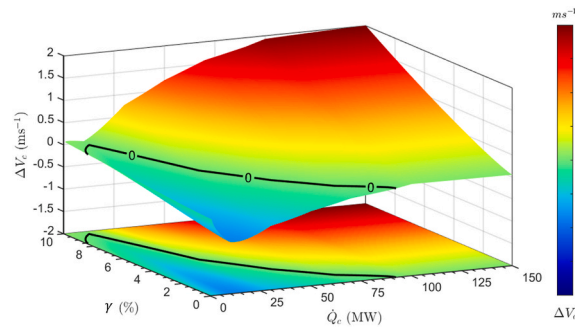


Fig. 9. The difference of V_c calculated based on the correlations by Li et al. [11] and by NFPA 502 [26]. ΔV_c is defined as the critical velocity calculated based on the correlation by Li et al. minus that calculated based on the correlation by NFPA.

5. Conclusion

In this work, the equations for the critical velocity, which are widely applied in designing the ventilation systems of tunnels, were theoretically solved in the full range of parameters. Specifically, our study includes:

- The equations governing the critical velocity were simplified to a cubic equation, which was proved to have only one positive real root. Hence, the positive real root of the simplified equation without constraints was determined to be the unique solution to the equations governing the critical velocity with constraints.
- The analytical expression of the critical velocity was derived, which was proved to be applicable for the full range of parameters considered.

In addition, the analytical solution was compared with the iterative, the simulation and the experimental results. What we found are summarized below:

- The analytical solution is confirmed by comparing with the iterative solution. There is no need to consider the constraints in the analytical solution while the constraints should be incorporated into the iterative solution somehow to avoid returning negative result for cases of high HRR and/or low tunnel height.
- The analytical solution was confirmed by experimental and simulation results. However, the analytical solution is highly related to the correlation of the Froude number. Thus, an appropriate correlation of the Froude number should be carefully selected.
- Two correlations of the Froude number, respectively from the NFPA 502 standard and the work by Li et al. were substituted into the analytical solution. For lower HRR and tunnel slope, the predicted critical velocity based on the correlation recommended by Li et al. is lower than that based on the one from the NFPA 502 standard.

Author statements

Yao Hong: Conceptualization, Investigation, Methodology, Writing - original draft, Formal analysis. **Ceji Fu:** Writing - review & editing, Funding acquisition, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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